

All for one and none forall: Compiling polymorphic relations without monomorphization

Dmitri Volkov, Yafei Yang, Chung-chieh Shan

2026 August 24



```

(defrel (graph
  (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 2))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 3))))

```

	to			
	0	1	2	3
from	0	0	1	0
	1	1	0	1
	2	0	0	0
	3	0	0	0

Can represent the graph as a relation or as an adjacency matrix.
 semiringKanren computes matrices for arbitrary relations!

semiringKanren values are drawn from algebraic data types.
Let x be a value of type α , and y be a value of type β .
We notate this as:

$$x : \alpha$$
$$y : \beta$$

semiringKanren has the following value constructors and corresponding types:

- ▶ `sole` : `Unit`
- ▶ `(left x)` : `(Sum  $\alpha$   $\beta$ )`
- ▶ `(right y)` : `(Sum  $\alpha$   $\beta$ )`

(semiringKanren also has product types)

We abbreviate types by their size, and index values.

$(\text{Sum Unit } (\text{Sum Unit Unit})) \mapsto 3$

$(\text{left sole}) \mapsto 0_3$

$(\text{right (left sole)}) \mapsto 1_3$

$(\text{right (right sole)}) \mapsto 2_3$

Relations denote arrays; entries correspond to argument values, and hold weights. Here, weight 1 denotes success and weight 0 denotes failure.

```
(defrel (R (x : 3))  
  (== x 03))
```

$$\begin{matrix} 0_3 & 1_3 & 2_3 \\ \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

(semiringKanren supports computing over weights drawn from any commutative semiring)

disj is addition.

```
(defrel (R (x : 3))  
  (disj  
    (== x 03)  
    (== x 23)))
```

$$\begin{array}{r} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \\ \begin{matrix} 0_3 & 1_3 & 2_3 \\ \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \end{matrix} \end{array}$$

conj is multiplication.

```
(defrel (R (x : 3))  
  (conj  
    (== x 03)  
    (== x 23)))
```

$$\begin{array}{ccc} [1 & 0 & 0] * [0 & 0 & 1] \\ & 0_3 & 1_3 & 2_3 \\ & [0 & 0 & 0] \end{array}$$

Multi-argument relations denote multidimensional arrays.

```
(defrel (R (x : 3) (y : 3)
  (== x y)))
```

$$\begin{array}{c} 0_3 \quad 1_3 \quad 2_3 \\ 0_3 \quad \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \\ 1_3 \quad \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \\ 2_3 \quad \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \end{array}$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

